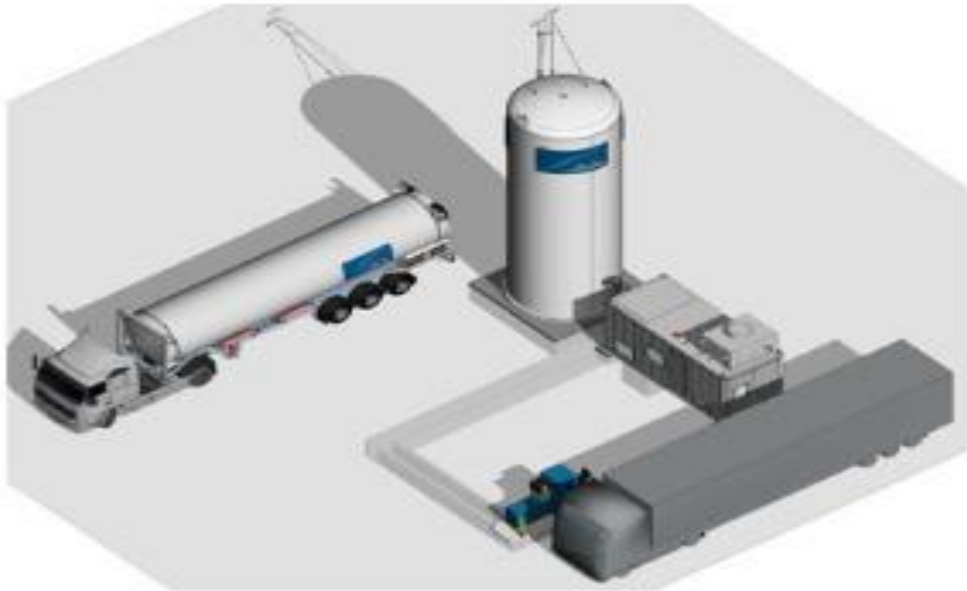


Pilot Unit for the production of Green Hydrogen In Mauritania





Introduction

- Green Hydrogen: a clean and renewable source of energy, with zero carbon emissions,
- In Mauritania, a pilot unit to produce Green Hydrogen will be installed, using renewable energy (PPA),
- Pilot plant characteristics:
 - 2000 kg per day production of gH₂,
 - Gathered data will be used for CMMZE's project of a 200 MW electrolysis capacity, to be built.



Green Hydrogen Production Process

- The key element in the process is the electrolyzer which splits the water into hydrogen and oxygen,
- The electrolyzer is fed with green electrons,
- The gH₂ could be used in different applications:
 - to replace “grey H₂”,
 - as fuel in transportation,
 - to produce green derivatives: gNH₃...
- The pilot unit will use a containerized 5 MW alkaline electrolyzer,
- Desalinated water will be provided by an existing desal. plant in the region.

Pilot Unit – First gH2 in Mauritania

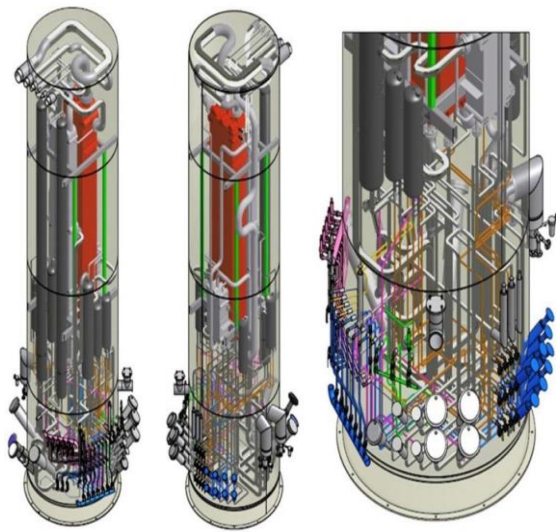
- Pilot unit's key figures:
 - 10,000 m2 land parcel,
 - 5MW alkaline electrolyser, 2000 kg/day,
 - 99.999% purity H2, up to 450 bar storage,
- Main targets are:
 - First green producing unit in Mauritania,
 - Opening the door to bigger projects in country,
 - Offering the opportunity to universities and engineering schools to conduct studies, research on gH2
 - Collaborating with industrials to assess/test the usage of gH2 in their processes to reduce carbon footprint.





CMMZE's Next Steps

- The pilot unit is our starting point Mauritania, it is also our starting point in the Africa:
 - Other pilot plants will follow in other countries: Morocco, Tunisia, UAE, Mauritania,
- The next step in Mauritania will be the 80 Ton per day gH2 project (200-MW electrolysis capacity)
- Based on renewable energy (PPA)
 - Building on the experience from the pilot unit, with Mauritania's local conditions
- Partnering with local industries to integrate gH2 in their processes, and comply with evolving export rules,
- Exploring gH2 and/or derivatives to Europe with already identified off-takers.





- **CMMZE**

- **CMM Zero Emission** is a company established in Tunisia to pursue the development of green hydrogen projects using electrolyzers to split water powered by Renewable Energy sources.
- www.cmmze.net
- **Address:** CMMZE S.A. Premium building B 4.3 Lac 2 Tunis, Tunisie